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Reg. No. | | | | | | | | | | | [17EE41]

FOURTH SEMESTER B.Tech. EXAMINATION, JUNE – 2019

ELECTROMAGNETIC FIELD THEORY

Time: 3 Hours **Maximum Marks:70**

Instructions:

i. Answer any **FIVE** full Questions choosing at least any two full questions from each part.

ii. Assume any missing data suitably if necessary

iii. Draw neat diagrams wherever required.

PART - A

1. a. State and explain Coulomb's law in vector form. **04**

   b. Using Gauss law, find the intensity of electric field at all points due to a charged concentric spherical shell of radius 'R' mts having a charge of 'Q' C distributed uniformly on its surface. Sketch the field variation with respect to distance. **06**

   c. Find ρ_v for r = 0.06m with respect to the field $\vec{D} = 5r^2\hat{a}_r\, mC/m^2$. **04**

2. a. Show that electric field intensity is equal to negative potential gradient. **04**

   b. Find the energy stored in the system of 4 charges each 4η C placed at the corners of a square of side 1m. **04**

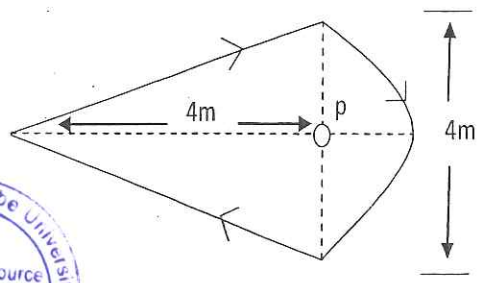
   c. Derive the relationship between tangential and normal components of $\bar{D}$ and $\bar{E}$ between dielectric - dielectric interface. **06**

3. a. Derive an expression for Poisson's and Laplace equation and express the same in all co-ordinate systems. **08**

   b. Applying Laplace equation, obtain an expression for electric field strength in Cartesian co-ordinate system between two parallel planes separated by a distance of 'd' m in the Z-direction having a potential difference applied between them. **06**

4. a. State and explain Biot Savert's law applied to magnetic fields. **04**

   b. Find the value of **magnetic flux density** at point 'P' for the current circuit carrying a current of 10A as shown below. **06**



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c. Derive an expression for Gauss Law in point form applied to magnetic fields. **04**

PART - B

5. a. Derive an expression for force between two parallel conductors carrying a current of I amps in opposite direction. **08**

b. A circular loop of radius 'r' mts and carries a current of 'I' amps lies in the Z-plane. Find the torque which results if the current is in a_ϕ direction and there is an uniform field given by $B = \frac{B_o}{\sqrt{2}}(\hat{a}_x + \hat{a}_z)T$. **06**

6. a. Derive an expression for statically and dynamically induced EMF. **06**

b. List the Maxwell's equation both in integral form and differential form applied to electric and magnetic field and also list for free space applications. **08**

7. a. Derive an expression for wave equations in terms of $\bar{D}$ and $\bar{E}$. **04**

b. Electric field intensity at a distance of 10 km from a radio station was found to be 2.2mV/m. Calculate the power density and total power radiated from radio station **06**

c. With the help of plot, explain the significance of skin depth. **04**

8. a. List the sources of electromagnetic fields. **04**

b. Enumerate the applications of electromagnetic waves in Bio medical science. **06**

c. For silver, conductivity $\sigma = 3 \times 10^6$ mho per m, at what frequency the depth of penetration be 1mm. **04**